

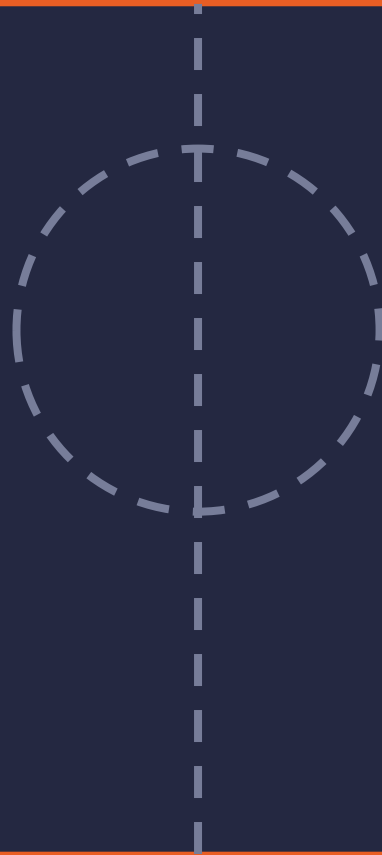
Deep Analysis of how NIL affects the Competitiveness of the League using WAB



By The Pennsylvania Suns

Project Overview

- NIL
- Purpose of Project
- Code Overview
- Data
- Problems



THE NIL

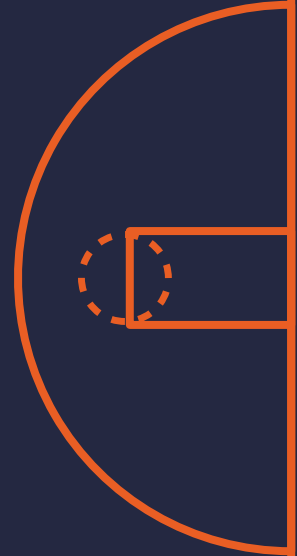
The NIL (Name, Image, and Likeness) added in 2021, refers to the rights of college athletes to earn compensation from endorsements and deals.





Purpose

- Show if the NIL had an impact throughout the years.
- Show the competitive gap throughout the years.
- Show if WAB or Quadrant wins are more impactful for strength of team.
- Predict next year of the league.



Set Up:

```
import pandas as pd
import requests
import notebook
import sqlalchemy
import psycopg2
import json
import numpy as np
from datetime import datetime
import cbbd
import glob
import os
from datetime import datetime, timedelta
from sqlalchemy import create_engine
```

Pull Keys from Json:

```
• with open("keys.json", "r") as f:  
    keys = json.load(f)  
    API_KEY = keys.get("API_KEY")
```

Making combined CSV:

```
if not os.path.exists("data/NCAA Statistics Combined.csv"):  
    files = glob.glob("data/NCAA Statistics *.csv")  
  
    dfs = []  
    for f in files:  
        temp = pd.read_csv(f)  
        temp["year"] = os.path.basename(f).replace("NCAA Statistics ", "").replace(".csv", "")  
        dfs.append(temp)  
  
    df = pd.concat(dfs, ignore_index=True)  
  
    # Save to a new CSV  
    df = df[["year"] + [col for col in df.columns if col != "year"]]  
    df.to_csv("data/NCAA Statistics Combined.csv", index=False)  
  
    print(f"Done! Combined {len(files)} files with {len(df)} total rows.")  
    print(df.head())  
else:  
    print("Combined file already exists.")
```

Web Calculation:

```
def win_prob_vs_bubble(opponent_elo: float, bubble_elo: float) -> float:
    diff = opponent_elo - bubble_elo
    prob = 1.0 / (1.0 + 10 ** (diff / 400))
    return float(np.clip(prob, 0.05, 0.95))
```

```
for season in seasons:
    print(f"\nFetching season {season}...")
    year1 = season - 1
    year2 = season
    feb_end = datetime(year2, 3, 1) - timedelta(days=1)

    with cbbd.ApiClient(configuration) as api_client:
        games_raw_1 = cbbd.GamesApi(api_client).get_games(
            start_date_range=datetime(year1, 11, 1),
            end_date_range=datetime(year1, 12, 31),
            status=cbbd.GameStatus.FINAL
        )
        games_raw_2 = cbbd.GamesApi(api_client).get_games(
            start_date_range=datetime(year2, 1, 1),
            end_date_range=feb_end,
            status=cbbd.GameStatus.FINAL
        )
        games_raw_3 = cbbd.GamesApi(api_client).get_games(
            start_date_range=datetime(year2, 3, 1),
            end_date_range=datetime(year2, 3, 14),
            status=cbbd.GameStatus.FINAL
        )
        elo_raw = cbbd.RatingsApi(api_client).get_elo(season=season)

    games_raw = games_raw_1 + games_raw_2 + games_raw_3
```

Final Dataframes:

```
# Final combined DataFrame - outside the season loop
WAB_data = (
    pd.DataFrame(all_results)
    .sort_values(["Season", "Wab"], ascending=[True, False])
    .reset_index(drop=True)
)
```

```
Predictions_Data = pd.read_csv("data/predictions_2027.csv")
```

```
NCAA_Data = pd.read_csv("data/NCAA Statistics Combined.csv")
```


Creating Engine:

```
PG_URL = 'postgresql+psycopg2://postgres:postgres@localhost:5440/postgres'  
engine = create_engine(PG_URL)  
print('Database:', PG_URL)
```

✓ 0.0s

Database: postgresql+psycopg2://postgres:postgres@localhost:5440/postgres

```
NCAA_Data.to_sql('NCAA', engine, if_exists='replace', index=False)
```

✓ 0.6s

512

```
WAB_data.to_sql('Wab', engine, if_exists='replace', index=False)
```

✓ 0.8s

303

```
Predictions_Data.to_sql('Predictions', engine, if_exists='replace', index=False)
```

✓ 0.2s

364

```
engine.dispose()  
print('Connection closed')
```

✓ 0.0s

Connection closed



<http://localhost:3000/public/dashboard/263eee58-3194-4181-8fc7-f098a0bf9685>

Problems

- Excel
- Columns and cleaning data
- WAB
- Engine Container/port
- Predictor
- Github merging/pulling

